CN Practical 6: Configure Static IP Address

Aim: To configure a static IP address in the operating system with an appropriate subnet mask for a given problem statement.

Theory:

An Internet Protocol (IP) address is a logical address assigned to a device connected to a computer network. It helps identify the device and enables communication with other devices. A static IP address is configured manually by the user or network administrator and does not change automatically. It is useful for devices such as servers, printers and network equipment that require a consistent address.

A subnet mask identifies the network portion and host portion of an IPv4 address. For example, the subnet mask 255.255.255.0 corresponds to /24, meaning that the first 24 bits identify the network. The default gateway is the router address used to communicate with devices outside the local network. A DNS server translates domain names into IP addresses.

Example configuration:

Parameter	Value
IP address	192.168.1.10
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
Preferred DNS	8.8.8.8

Procedure:

1. Open Control Panel in Windows.
2. Select Network and Sharing Center.
3. Click Change adapter settings.
4. Right-click the Ethernet adapter and select Properties.
5. Select Internet Protocol Version 4 (TCP/IPv4).
6. Click Properties.
7. Select “Use the following IP address.”
8. Enter the IP address, subnet mask and default gateway.
9. Enter the preferred DNS server if required.
10. Click OK to save the settings.
11. Open Command Prompt and type ipconfig.
12. Verify that the static IP address has been configured correctly.

Observation:

The static IP address and subnet mask were entered manually in the network settings. The ipconfig command displayed the configured IPv4 address and other network parameters. The device was ready to communicate with other devices in the same subnet, provided the addresses and physical connections were correct.

Conclusion:

A static IP address was configured successfully in the Windows operating system. The role of the IP address, subnet mask, default gateway and DNS server was studied. The configuration was verified using the ipconfig command.